Automated monthly sales reporting

Portfolio case study | Python, SQLite and Excel | Kaggle Superstore sample

The recurring task

A business receives a new sales CSV every month. It needs a consistent workbook, traceable totals and saved review notes, without rebuilding formulas or overwriting the last usable report when a file is malformed.

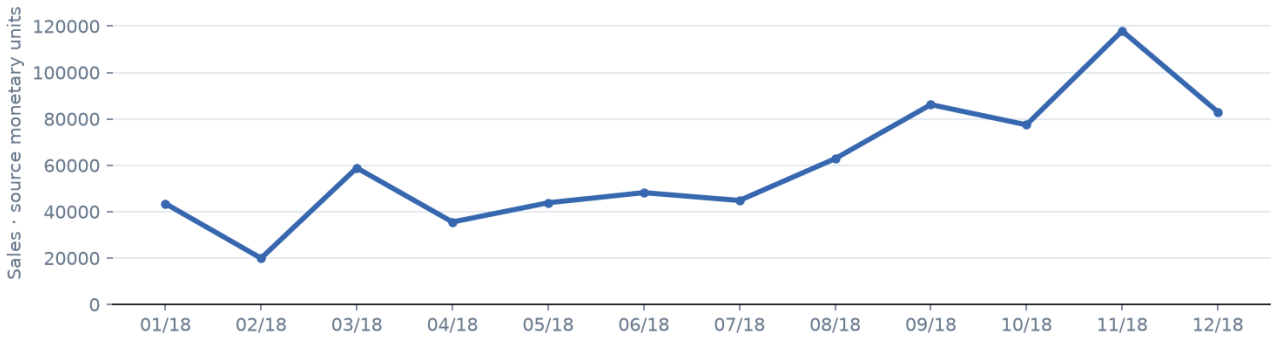
The delivered workflow

Place files named sales_YYYY-MM.csv into the input folder and run **python refresh.py**. The program validates IDs, dates and amounts, builds SQL aggregates, generates the Excel workbook and writes a versioned run log. It updates latest.json only after the workbook has been generated successfully.

Demonstrated with actual workbook refreshes

REFRESH STEP	FILES	LINES	OBSERVED SALES
Before December delivery	47	9,341	2,178,506.3939
After December delivery	48	9,800	2,261,536.7827
Repeat the same refresh	48	9,800	2,261,536.7827

The December delivery adds 459 lines and 83,030.3888 source monetary units. A temporary, clearly labeled review note remains attached to its November/category key. Repeating the same input leaves totals unchanged. A conflicting Row ID then blocks publication while the previous pointer and workbook remain available.



This replays 48 historical partitions from one downloaded sample dataset. It does not claim that Kaggle publishes a new monthly file or that the sample represents a current retailer.

Controls, handoff & limitations

A practical example for recurring spreadsheet and data-cleaning contracts

What the workbook contains

Monthly report: the latest period, sales, distinct orders, average order amount, year-over-year comparison and a native trend chart. **Monthly detail:** all observed periods, imported counts, prior-period references and live ratio formulas. **Category review:** editable owner, status and note cells linked to stable month/category keys.

Notes and refresh behavior

Export Category review as CSV UTF-8 and run **python src/import_notes.py exported-review.csv**. The importer merges annotations, keeps older keys and ignores untouched template rows. Reviewed requires both owner and note. Refreshing reads the separate notes input; it does not overwrite it. Keys absent from the current input appear in unmatched_notes.csv.

Validation and publication

Identical repeated Row IDs are skipped; conflicting IDs, missing headers, invalid dates, dates in the wrong filename month, inconsistent order dates and unsupported amounts stop the run. Every accepted row retains its source filename and row number. A missing month is unavailable; an explicit header-only monthly file is known zero. Six controls reconcile source rows and SQL totals. Each successful version contains its input hashes, metrics and workbook checks.

Evidence

Twenty-three automated tests cover the ingestion, note-import and publication rules. Ten real-engine replay checks cover a new month, note retention, a repeat run and a rejected conflict. Independent original-source and XLSX checks confirm the sales totals, typed dates, year-over-year formulas, chart bindings, dropdown and empty delivered notes.

Source and operating limits

Downloaded Kaggle dataset: **rohitsahoo/sales-forecasting**. Its data card labels the dataset GPL 2. The project records the exact source SHA-256 and supplies selected original columns as reproducible monthly inputs. The Superstore sample covers Jan 3, 2015–Dec 30, 2018. Currency is unspecified by the mirror, so the report preserves source monetary units and four-decimal precision. Displayed amounts use two decimals.

Runtime and review

Core ingestion, SQL reporting and tests use standard-library Python. Excel generation requires Node.js and the compatible artifact-tool authoring runtime documented in SETUP.md. The delivered workbook is ready to inspect. Formula recalculation was tested in that engine; native desktop Excel was not available. Worksheet edits require CSV note import before refresh; there is no background watcher or automatic workbook write-back.

Kaggle dataset and license label | Complete source, test evidence and refresh guide accompany this briefing.